Oriented Object Detection for Remote Sensing Images via Object-Wise Rotation-Invariant Semantic Representation

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***Abstract*— Oriented object detection (OOD) in remote sensing images (RSIs) remains a challenging work due to an arbitrary orientation of instances. Learning rotation-invariant features is critical in modeling a fixed descriptor for instances with its rotated variants. However, most existing methods construct the descriptor from the perspectives of data or feature augmentation, but ignore the exploration of potentially useful supervision information inside the detection algorithm. In this article, we propose an object-wise rotation-invariant semantic representation (ORSR) framework, which synergizes the exploration of latent supervision, rotation-invariant learning, and guided attention mechanism into a unified network to boost the performance of OOD in RSIs. First, supervised by our constructed pseudo-ground truth of segmentation masks, a semantic segmentation branch is built along with the detection algorithm to refine the representation of backbone features. Moreover, a consistency loss function is proposed to encourage the segmentation branch to make fixed predictions for backbone features with its rotated variants. Considering that segmentation predictions remain the same affine transformations before and after rotating, we further construct a Kullback–Leibler (KL) divergence-based similarity loss function to encourage the network to model the rotation-invariant features. Finally, we separate the "object" descriptor from the segmentation predictions to extend the implicit constraint in our proposed semantic segmentation branch. The separated "object" descriptor not only involves the spatial regularizer to emphasize the high-responsive regions in the image but also can be guided by the constructed consistency loss function. We evaluate our proposed ORSR on the challenging DOTA, DIOR-R, and HRSC2016 datasets. Extensive experiments demonstrate that the proposed ORSR achieves competitive performance compared to other single-scale and multiscale (MS) detection methods.**

***Index Terms*— Oriented object detection (OOD), remote sensing images (RSIs), rotation-invariant learning, semantic representation.**

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I. Introduction

DUE to the advances in deep learning, the convolutional neural network (CNN) has facilitated the development of multiple computer visual tasks [1], [2], including image classification [3], [4], [5], object detection [6], [7], [8], [9], [10], [11], and so on. However, different from objects in natural scenes [12], [13], [14], aerial instances often own many specific characteristics, such as arbitrary orientation, dense distribution, and huge scale variation. Thus, oriented object detection (OOD) [15], [16], [17], [18], [19], [20] in remote sensing images (RSIs) has emerged as a crucial, yet challenging problem in recent years.

Many OOD detectors [21], [22], [23], [24] have benefited from the Faster Region-CNN (Faster R-CNN) framework [25]. These methods transferred a horizontal region of interest (HRoI) to the oriented one for better perceiving objects with different angles and aspect ratios via the well-designed warping modules. The aforementioned methods explicitly reduce the number of useless background pixels in anchors or candidate bounding boxes. Correspondingly, spatial transformer [26] and deformable convolution [27] had been proposed to implicitly model the geometric variations by feature selection. Considering that the original CNN exhibits limited ability to model rotation-invariant features, rotated RoI Pooling [28] was devoted to warping region features precisely in the 2-D planar. These RoI warping approaches have become the basic blocks in the OOD algorithm to improve the OOD performance in RSIs. However, without a larger capacity network and ample training samples, regular CNN features cannot precisely model rotation-invariant variation. To address these limitations, the rotation-equivariant detector (ReDet) incorporated group convolution [29] into the object detector, which simultaneously encoded both rotation equivariance and invariance in CNNs.

Despite excellent progress has been achieved in OOD, all the aforementioned approaches do not explore the potentially

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useful supervision information in the detection algorithm. Recently, jointly training multiple tasks can excavate the complementary benefits between different branches, which offers great potential to boost many computer vision tasks. Mask R-CNN [30] added a parallel segmentation branch in the Faster R-CNN algorithm to predict instance masks. Both detection and segmentation branches benefit from a unified loss function. OWSR [31] improved the region proposal network (RPN) with semantic segmentation tasks to suppress background clutter as much as possible. The masked rotate regions of ground truth were regarded as the additional supervision for the segmentation branch. The mainstream self-training frameworks [32], [33], [34] iteratively assign extra annotations to the labeled or unlabeled samples. These knowledge distillation strategies pose great potential to explore the additional supervision for generating more reliable training samples.

In this article, we propose an object-wise rotation-invariant semantic representation (ORSR) network for OOD in RSIs. Our ORSR unifies the exploration of latent supervision, rotation-invariant learning, and guided attention mechanism into a complete framework, which provides a straightforward fashion to explore the interaction among these modules. First, we mask the rotated bounding boxes in the image to generate segmentation masks and build a segmentation branch along with the detection backbone. A top-down feature fusion pipeline is introduced to make the best compromise between high-level semantics and detailed spatial (low-level) information. Moreover, we propose a consistency loss to encourage the segmentation branch to make the fixed predictions for the fused features with its rotated variants. The aforementioned strategy promotes our detection network to effectively learn the rotation-invariant image features. Considering that the predicted 2-D segmentation mask involves "nonobject" and "object" descriptors, the latter can act as a spatial attention map to emphasize the high-responsive regions in features. Surprisingly, the "object" descriptor not only involves the spatial regularizer but also can be supervised by our proposed consistency loss function. Extensive experiments on DOTA, DIOR-R, and HRSC2016 datasets demonstrate that our ORSR achieves competitive performance among many single-scale and multiscale (MS) OOD methods.

The remaining sections of this article are structured as follows. Section II provides a review of related methods. In Section III, we present the detailed descriptions of our ORSR. The performance results achieved by our method, along with the comparisons against various state-of-the-art approaches are presented in Section IV. Finally, Section V concludes this article with final remarks and provides directions for further research.

II. Related Works

A. OOD in RSIs

Due to arbitrary orientation, dense distribution, and large aspect ratio of targets in RSIs, object detection in remote sensing scenarios is more challenging compared to natural scenes. To better cover the instances and reduce redundant pixels, OOD is the preferred choice for detecting instances in RSIs. Five-parameter and eight-parameter regression methods are two typical approaches to defining an oriented bounding box (OBB). However, due to inappropriate angular ordering and angular periodicity, the aforementioned methods suffer from the boundary discontinuity problem. To alleviate the aforementioned limitation, the circular smooth label (CSL) and densely coded label (DCL) modules [35] were proposed to predict the rotational orientation of objects in a classification fashion. Removing boundary discontinuity caused by angle regression, DCL further reduced the excessive computation cost of CSL. Sparse label assignment (SLA) [36] was constructed to balance the inconsistencies between the category classification and bounding box regression tasks. The proper label assignment strategy ensured accurate and meaningful interpretation of instances in RPN, which posed great potential for excavating the most high-quality sparse anchors. Many methods focus on emphasizing all the discriminative regions in images [37], [38], [39], [40]. Zhang et al. [37] proposed a context-aware detection network (CAD-Net) to model the spatial correlation among different oriented objects at the image level and object level. The global and local contextual spatial information of target instances were captured by the constructed MS attention-enhanced module to better locate the position of objects. Ming et al. [39] constructed two separated subnetworks to simultaneously perform category classification and bounding box regression tasks. The decoupled features were adopted to progressively refine the performance of OOD. A top-down pathway was constructed in [40] to integrate features of the shallow layer with high-semantic contextual representations. Features at different levels were merged by the proposed feature-fusion module to obtain more powerful representations of MS instances.

B. Rotation-Invariant Learning

Recent methods tend to model rotation-invariant features to detect arbitrarily oriented objects in RSIs. Exploring unified representations of instances with their rotated variants is one of the most commonly used strategies to model rotation-invariant features. RoI transformer [28] proposed a learnable network to transfer the HRoI into the rotated one, which effectively eliminated the mismatches between common horizontal and special orientated bounding boxes. Oriented RPN (O-RPN) [41] generated high-quality oriented anchors to better cover the target objects. Moreover, O-RPN constructed a midpoint offset representation module (MPRM) to further decrease the risk of overfitting caused by the frequent offset transformation. Cheng et al. [42] imposed an explicit regularization constraint on their proposed rotation-invariant convolutional layer, which enforced image features to have similar high-order mappings with its multiple rotated ones. Ma et al. [43] extended RoI warping to the rotated RoI warping, aiming for generating inclined candidate proposals with object-orientation angles. The angle predictions were further introduced in the bounding box regression branch to fit the proposals into appropriate instance regions. A ReDet [29] modeled the rotation-equivariant features by incorporating the rotation-equivariant network into the common detector.

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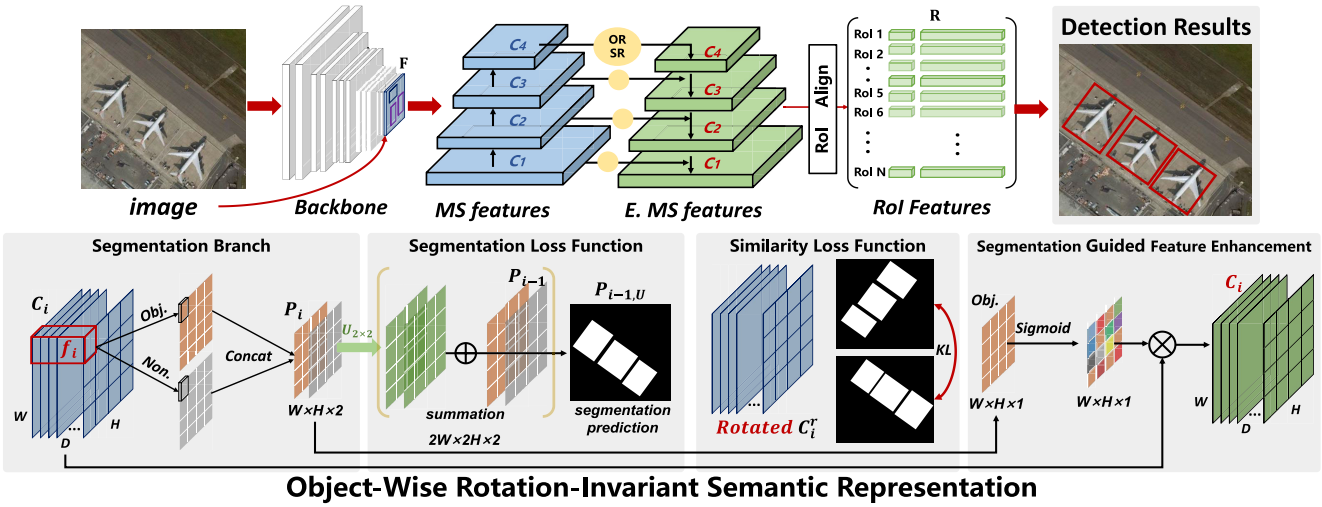


Fig. 1. Overview of our ORSR network, which consists of a segmentation branch, rotation-invariant learning branch, and a GFE branch. After feeding images into the backbone network, the output MS features are used to predict the segmentation maps. Then, the constructed consistency loss function encourages the segmentation branch to make fixed predictions for fused features with its rotated variants. Finally, we separate the "object" descriptor from the segmentation prediction to highlight the relevant regions and suppress the redundant ones.

The rotation-invariant RoI Align was also proposed to adaptively excavate rotation-invariant features according to the predicted orientation of RoIs. Methods [15], [19], [21] were all devoted to merging frequency-domain analysis tools into the CNN to model rotation-invariant features. The unified spatial-frequency descriptors were proposed to better describe oriented objects, which can apply to arbitrary angles of rotated instances.

C. Exploring of Extra Supervision

Exploring additional supervision to jointly train the constructed multitasks network for leveraging the complementary advantages among different tasks has gained significant attention in many computer vision tasks. OWSR [31] unified a semantic segmentation task into the detection framework, which promoted the RPN branch to suppress background clutter as much as possible. However, due to the arbitrary orientation of targets in RSIs, extra segmentation supervisions under a single-direction constraint suffer from modeling rotation-invariant image features. Li et al. [34] proposed a pseudo-segmentation supervision to guide the parameter optimization in their attention module. The attention mining loss (AML) function was constructed to force that regions outside the attention map are irrelevant to the object recognition task. However, the aforementioned AML overlooks the exploration of potential supervisions within the network itself. To emphasize the response of small objects in RSIs, Wang et al. [38] proposed an area-weighted loss function to guide their constructed feature-merged single-shot detection (FMSSD) framework. Both MS and the same scale of features were utilized to model high-semantic contextual representations. Yang et al. [44] proposed a multilabel classification loss (MLCL)-guided attention module to enhance the relevant regions and suppress the redundant ones. Supervised by the constructed MLCL, attention maps act as a spatial regularizer, aiming to emphasize the discriminative ability of the framework as well as detection accuracy. Object-wise point-guided localization detector (OPLD) [45] unified a key-point estimation task into the detection network to alleviate the discontinuous bounding boxes regression problem. However, instances in RSIs are densely distributed. The center points of adjacent objects are close, resulting in missing detection for OOD.

III. Proposed Method

A. Overview

The overview of our detection network is illustrated in Fig. 1. As defined in [46], the ResNet backbone feature extractor is used to attain MS image features $C_i \in \mathbb{R}^{W_i \times H_i \times D_i}$, where W_i, H_i, and D_i denote the width, height, and channel of C_i, respectively. Then, C_i with its rotated variants are fed into the proposed object-wise rotation-invariant semantic segmentation network, which consists of a segmentation branch, a rotation-invariant learning branch, and a guided attention mechanism. The segmentation branch generates pseudo-segmentation masks via masking the rotated bounding boxes in images and then encourages the segmentation branch to make accurate predictions for C_i and C_i^r. Moreover, we propose a consistency loss function guided rotation-invariant learning branch, which aims to supervise the constructed segmentation network. Importantly, pseudo-segmentation annotations remain the same affine transformation in predictions before and after rotation. We further construct a Kullback–Leibler (KL) divergence-based similarity loss function to encourage that the segmentation results before and after rotation should be consistent. Considering that the 2-D segmentation prediction involves "object" and "nonobject" feature descriptors, where the former explicitly indicates whether the pixel regions contain latent instances. Therefore, we propose a guided feature enhancement (GFE) branch to further emphasize the relevant regions and suppress the redundant ones. Finally, all the enhanced MS features are fed into the RoI Align, category classification, and bounding

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box regression networks to detect the oriented objects in images.

B. Segmentation Branch

Different from the PASCAL VOC dataset, which contains both detection and segmentation annotations, mainstream OOD datasets only have image-level and bounding box labels. Thus, we use the rotated bounding boxes $B = [x, y, w, h]$ to generate the segmentation masks Y. Specifically, a masking operation is performed to set the pixels within the bounding boxes to 1, while pixels outside the bounding boxes are set to 0. Therefore, the element-wise values in Y belong to [0, 1].

Given an input image $I \in \mathbb{R}^{W\times H\times 3}$, where W and H denote the width and height of I, respectively. Four scales of image features are extracted by the ResNet backbone network: $C_1 \in \mathbb{R}^{(W/4)\times(H/4)\times 256}$, $C_2 \in \mathbb{R}^{(W/8)\times(H/8)\times 512}$, $C_3 \in \mathbb{R}^{(W/16)\times(H/16)\times 1024}$, and $C_4 \in \mathbb{R}^{(W/32)\times(H/32)\times 2048}$. Then, we down-sample the generated segmentation mask to achieve different scale of GT masks $Y_0 = Y \in \mathbb{R}^{W\times H}$, $Y_1 \in \mathbb{R}^{(W/4)\times(H/4)}$, $Y_2 \in \mathbb{R}^{(W/8)\times(H/8)}$, and $Y_3 \in \mathbb{R}^{(W/16)\times(H/16)}$.

Considering that objects in RSIs often have different shapes or scales, fusing MS image features (C_1, C_2, C_3, C_4) is an intuitive solution to promote the representation ability of the detection network. Concretely, we use four different 1×1 convolution layers $f_{i,1\times1}|_{i=1}^{4}$ to predict object or nonobject (background) segmentation predictions, which can be formulated as

$$P_i = \mu\big(f_{i,1\times1}(C_i)\big), \quad i = 1, 2, 3, 4 \tag{1}$$

where μ represents the ReLU activation layer. $P_1 \in \mathbb{R}^{(W/4)\times(H/4)\times 2}$, $P_2 \in \mathbb{R}^{(W/8)\times(H/8)\times 2}$, $P_3 \in \mathbb{R}^{(W/16)\times(H/16)\times 2}$, and $P_4 \in \mathbb{R}^{(W/32)\times(H/32)\times 2}$ are different coarse segmentation maps. Note that (1) does not change the spatial scale of each input feature, and the down-sampling rate between P_i and P_{i-1} is 2. Moreover, we construct a top-down feature fusion pipeline to combine coarse, shallow layer information with fine, deep layer features

$$P_{i-1,U} = U_{i,2\times2}(P_i) + P_{i-1}, \quad i = 4, 3, 2 \tag{2}$$

where $U_{i,2\times2}$ denotes the deconvolutional layer with kernel size is 2. Finally, an up-sampling layer $U_{4\times4}$ is adopted for resizing $P_{1,U}$ to match the scale of segmentation map Y

$$P_{0,U} = f_{1\times1}\big(U_{4\times4}\big(P_{1,U}\big)\big) \tag{3}$$

where $f_{1\times1}$ denotes a 1×1 convolutional layer. In brief, the aforementioned operations achieve four pairs of predictions and GT: $(P_{0,U}$ & $Y_0)$, $(P_{1,U}$ & $Y_1)$, $(P_{2,U}$ & $Y_2)$, and $(P_{3,U}$ & $Y_3)$.

Here, we use the rotation affine transformation $\mathcal{T}$ to produce different rotated features and GT segmentation maps as

$$\mathcal{T}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \end{bmatrix} \tag{4}$$

$$C^r = \mathrm{GS}(C, \mathcal{T}) \tag{5}$$

$$Y^r = \mathrm{GS}(Y, \mathcal{T}) \tag{6}$$

where θ represents the rotation angle. GS denotes grid sample operation, which is defined in [26]. The predicted segmentation maps P_U^r can be achieved via feeding C^r into (1)–(3). Surprisingly, the aforementioned transformations will not change the information that the input features C focus on. All the segmentation predictions with the corresponding GT segmentation maps are sent to the rotation-invariant learning branch for supervising the generation of rotation-invariant features.

C. Rotation-Invariant Learning Branch

Rotation transformation is a widely used data augmentation for OOD in RSIs. As illustrated in Fig. 2, it is well known that instances with its rotated variants possess the same category, which also concludes that pseudo-segmentation annotations obtained by our method also remain the same affine transformation in image after rotation. Hence, we propose a consistency loss to supervise the aforementioned model optimization.

First, given the multiple pairs of predictions and GT, the standard cross-entropy loss function is used to supervise a pseudo-semantic segmentation task as

$$\mathcal{L}_{\mathrm{seg}} = \frac{1}{WH}\sum_{i=0}^{3}\sum_{t=1}^{W}\sum_{j=1}^{H} Y_{i,k,tj}\log\big(P_{i,U,k,tj}\big) + \frac{1}{WH}\sum_{i=0}^{3}\sum_{t=1}^{W}\sum_{j=1}^{H} Y^r_{i,k,tj}\log\big(P^r_{i,U,k,tj}\big) \tag{7}$$

where k denotes the kth semantic category. i is the number of segmentation predictions $P_{i,U}$, which is formulated in (1)–(3). W and H represent the width and height of ground truth Y and P, respectively. (t, j) represents the pixel position in features.

Remarks: Ideally, the predicted segmentation maps P and P^r remain the same affine transformation in the image after rotation, because C_r are obtained by rotating C. Thus, we further construct a KL divergence-based similarity loss function as

$$\mathcal{L}_{\mathrm{sim}} = D_{\mathrm{KL}}\big(P_U \| P_U^r\big) = \frac{1}{WH}\sum_{i=0}^{3}\sum_{t=1}^{W}\sum_{j=1}^{H} P_{i,U,tj}(x)\log\frac{P_{i,U,tj}(x)}{P^r_{i,U,tj}(x)}. \tag{8}$$

Importantly, we use an inverse rotation to guarantee P_U^r is rotated in the same direction as the original prediction P_U. Finally, the proposed consistency loss function $\mathcal{L}_{\mathrm{con}}$ is the summation of the cross-entropy loss and the proposed similarity loss functions as

$$\mathcal{L}_{\mathrm{con}} = \alpha_1\mathcal{L}_{\mathrm{seg}} + \alpha_2\mathcal{L}_{\mathrm{sim}} \tag{9}$$

where α_1 and α_2 are weighting parameters which are all set to 1.0 in our experiments.

D. GFE Branch

Although driving the segmentation branch to make the same predictions for image features before and after the rotation realizes rotation-invariant learning, such a solution ignores

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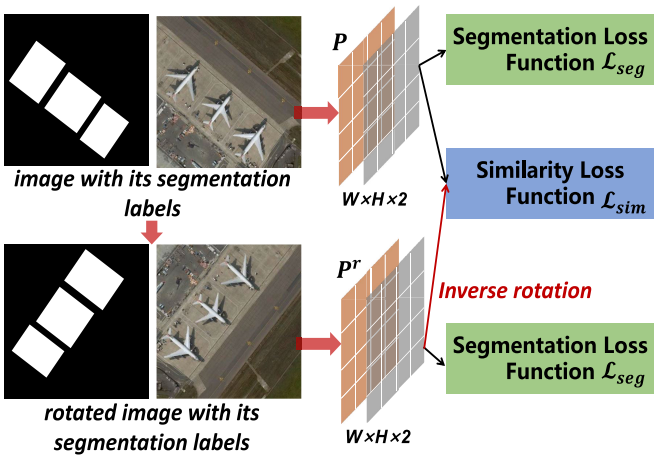


Fig. 2. Pipeline of our segmentation branch. P and P^r denote the segmentation predictions of original features with their rotated variants.

the fact that the "object" dimension in 2-D segmentation prediction explicitly indicates the latent instances in images. To alleviate this limitation, we further extend the implicit constraint in our segmentation branch to excavate all latent instances without the need for additional hyperparameters.

Considering that the 2-D segmentation predictions involve "object" and "nonobject" feature descriptors, we first separate the "object" descriptor from the corresponding dimension of segmentation predictions as

$$P_{i,U}^{\text{obj}} = P_{i,U}^{C=1}, \quad i = 1, 2, 3 \tag{10}$$

$$P_{4,U}^{\text{obj}} = f_{1\times1}\left(D_{2\times2}\left(P_{3,U}^{\text{obj}}\right)\right) \tag{11}$$

where C denotes the channel dimension of $P_{i,U}$ and $D_{2\times2}$ represents down-sampling operation. $f_{1\times1}$ denotes a 1×1 convolutional layer. $P_{1,U}^{\text{obj}} \in \mathbb{R}^{(W/4)\times(H/4)\times1}$, $P_{2,U}^{\text{obj}} \in \mathbb{R}^{(W/8)\times(H/8)\times1}$, $P_{3,U}^{\text{obj}} \in \mathbb{R}^{(W/16)\times(H/16)\times1}$, and $P_{4,U}^{\text{obj}} \in \mathbb{R}^{(W/32)\times(H/32)\times1}$ record the latent spatial responses of instances in image.

Remarks: Common feature enhancement module, attention mechanism, uses multiple convolutional and down–up-sampling layers to generate the spatial regularizer for highlighting the high-responsive regions. These regions always contain the fragments of certain instances. Surprisingly, "object" descriptor P^{obj} generated via (10)–(11) not only involves the similar spatial regularizer but also can be supervised by our proposed consistency loss function.

Here, we use the Sigmoid function σ to generate MS spatial attention masks as

$$A_i = \sigma\left(P_{i,U}^{\text{obj}}\right) \tag{12}$$

where the attention masks $A_i|_{i=1}^4$ have the same spatial scale as backbone features $C_i|_{i=1}^4$. Then, the enhanced backbone features can be formulated as

$$C_i^e = A_i \cdot C_i + C_i. \tag{13}$$

Finally, C_i^e are fed into the RoI Align, category classification, and bounding box regression networks to detect the oriented instances.

After unifying our proposed segmentation branch, rotation-invariant learning branch, and GFE branch into the detection network, the whole algorithm is optimized by

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{det}} + \lambda_2\mathcal{L}_{\text{cls}} + \lambda_3\mathcal{L}_{\text{con}} \tag{14}$$

where $\mathcal{L}_{\text{det}}$ denotes smooth L1 loss, which is defined in [25], $\mathcal{L}_{\text{cls}}$ represents multiclass cross-entropy loss which is supervised by the image-level classification annotations and λ_1, λ_2, and λ_3 are weighting parameters to balance these three loss functions. We set λ_1 and λ_2 to 1.0 and λ_3 to 0.1 for all experiments. The pipeline of our proposed ORSR is illustrated in Algorithm 1.

Algorithm 1 Object-Wise Rotation-Invariant Semantic Representation

Require: Input backbone features $C_i|_{i=1}^4$, GT of bounding boxes B, rotated angle θ
Ensure: Enhanced backbone features $C_i^e|_{i=1}^4$
Generate pseudo segmentation mask Y
Generate multi-scale pseudo segmentation masks $Y|_{s=0}^3$.
Generate rotated backbone features C_i^r and segmentation masks $Y^r|_{s=0}^3$ via Eqs. 4-6
Generate segmentation predictions P_i via Eq. 1
for $i = 4; i - 1; 2$ **do**
    Update $P_{i-1,U}$ by Eq. 2
Generate segmentation predictions $P_{0,U}$ via Eq. 3
for $i = 0 : 3$ **do**
    Calculate segmentation loss between $P_{i,U}$ and Y_i
Generate the rotated segmentation predictions $P_{i,U}^r$ via steps 4-7
for $i =$ 0:3 **do**
    Calculate segmentation loss between $P_{i,U}^r$ and Y_i^r
Calculate similarity loss between $P_{i,U}$ and $P_{i,U}^r$
Calculate consistency loss via Eq. 9
for $i =$ 1:4 **do**
    Generate the spatial regularizer $P_{i,U}^{\text{obj}}$ via Eqs. 10-11
    Generate the attention mask A_i via Eq. 12
    Generate enhanced backbone features C_i^e via Eq. 13
Feed the enhanced backbone features $C_i^e|_{i=1}^4$ into RoI Align layer and classification and regression branches to detect the oriented instances

IV. Experiments

Our experiments are implemented on the PyTorch platform with two blocks of 11 GB memory GeForce GTX 1080ti GPUs and evaluated on DOTA [47], DIOR-R [53], and HRSC2016 [60] datasets. A total of 25 approaches are compared in our experiments, including Faster RCNN-Oriented (FR-O) [47], RetineNet-Oriented (RTN-O) [61], improving bounding box representation (IBBR) [55], anchor-free oriented proposal generator (AOPG) [53], dual-aligned oriented detector (DODet) [54], image cascade network (ICN) [48], context-aware network (CADNet) [37], RoI Transformer [28], dynamic refinement network (DRN) [49], center probability map (CenterMap) [50], small cluttered and rotated detector (SCRDet) [51], feature-attentioned object detector (FAOD) [52], refine single-stage detector (R^3Det) [16], single-shot

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alignment network (S^2A-Net) [18], OPLD [45], objectness activation network (OAN) [57], ReDet [29], Oriented R-CNN [41], object relation module (ORM) [56], frequency-domain orientation learning (FDOL) [15], Gliding Vertex [17], object-wise semantic representation (OWSR) [31], Gaussian Wasserstein distance (GWD) [58], dynamic enhancement anchor network (DEA-Net) [59], and spatial-frequency feature fusion detector (SFFD) [19]. Most of the aforementioned methods have been published in recent years.

A. Datasets and Protocols

1) DOTA: DOTA is one of the most frequently used datasets for OOD in aerial images. It contains 2806 samples with image size ranges from 800 × 800 to 4000 × 4000. A total of 1 88 282 instances among 15 categories are annotated in DOTA: Plane (PL), Baseball diamond (BD), Bridge (BR), Ground track field (GTF), Small vehicle (SV), Large vehicle (LV), Ship (SH), Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), Harbor (HA), Swimming pool (SP), and Helicopter (HC). Following the standard settings in method [47], both training and validation sets are used for training. Moreover, each original image is cropped into 1024 × 1024 patches with a stride of 824 pixels. We adopt random horizontal flipping, without any other tricks, to train our models to avoid overfitting. To facilitate fair comparisons, we also use MS data augmentation strategy: each data is resized at three scales $\{0.5, 1.0, 1.5\}$ and random rotation $\{90°, 180°, 270°\}$ is also adopted for training and testing.

2) DIOR-R: The DIOR-R dataset is proposed in method [53], which has the same images as the original DIOR dataset [62]. After annotating OBBs for each sample, a total of 23 463 images with 1 92 518 instances are labeled in DIOR-R. Categories in DIOR-R include Airplane (APL), Baseball Field (BF), Basketball Court (BC), Bridge (BR), Chimney (CH), Airport (APO), Dam (DAM), Expressway Service Area (ESA), Expressway Toll Station (ETS), Harbor (HA), Overpass (OP), Ship (SH), Stadium (STA), Storage Tank (STO), Tennis Court (TC), Train Station (TS), Vehicle (VE) and Windmill (WM), Golf Field (GF), and Ground Track Field (GTF). Followed by the standard settings as [53], 11 725 images are used for training and the rest 11 738 samples are used for testing. The data augmentation strategies for DIOR-R are the same as the DOTA dataset.

3) HRSC2016: HRSC2016 is a popular dataset for oriented ship detection in RSIs. It contains 1061 images with size range from 300 × 300 to 1500 × 900 pixels. The training set has 436 images and the validation set contains 181 samples. The rest 444 images are used for testing. Following the standard settings as [60], all the images are resized to (800 and 512) while keeping the original aspect ratio of each image. Similar to the DOTA dataset, only horizontal flipping data augmentation is applied during training.

B. Implementation Details

Our method follows the pipeline of a two-stage detection algorithm. ReResNet with ReFPN [29] is adopted as the backbone network in our method. After attaining four different scales of image features, we set 15 anchors per location at each pyramid level for the RPN to identify object or nonobject regions. For the DOTA and DIOR-R datasets, we train our model for 12 epochs with the batch size being 2 on two GeForce GTX 1080ti GPUs. Following the learning rate linear transformation criterion in mmdetection [63], the SGD optimizer is used with the learning rate set to 0.005. We divide the learning rate by 10 at epoch 8 and epoch 11. The momentum and weight decay are set to 0.9 and 0.0001, respectively. For the HRSC2016 dataset, we train our method for 36 epochs and divide the learning rate by 10 at epochs 24 and 33.

C. Comparisons With the State-of-the-Art

1) Comparisons on DOTA: We list the detection performance of different single-scale OOD methods on the DOTA test set in Table I. Most of the compared approaches have been published in recent years. SFFD [19] unifies the frequency-domain analysis into the CNN-based detection framework and then excavates the interaction correlation among frequency and CNN instance-level features, which achieves 78.01% mAP. However, the channel dimension of instance-level features can reach thousands. Modeling the correlation of instance-level features is extremely time-consuming. Importantly, our proposed ORSR outperforms the FR-O [47], ICN [48], CADNet [37], RoI Transformer [28], DRN [49], CenterMap [50], SCRDet [51], FAOD [52], R^3Det [16], S^2A-Net [18], OPLD [45], AOPG [53], DODet [54], ReDet [29], IBBR [55], Oriented R-CNN [41], ORM [56], OAN [57], FDOL [15], and SFFD [19] by 23.90% (78.03% versus 54.13%), 9.87% (78.03% versus 68.16%), 8.13% (78.03% versus 69.90%), 8.47% (78.03% versus 69.56%), 7.33% (78.03% versus 70.70%), 6.29% (78.03% versus 71.74%), 5.42% (78.03% versus 72.61%), 4.75% (78.03% versus 73.28%), 4.29% (78.03% versus 73.74%), 3.91% (78.03% versus 74.12%), 3.85% (78.03% versus 74.18%), 2.64% (78.03% versus 75.39%), 2.14% (78.03% versus 75.89%), 1.78% (78.03% versus 76.25%), 1.78% (78.03% versus 76.25%), 1.75% (78.03% versus 76.28%), 1.30% (78.03% versus 76.73%), 1.30% (78.03% versus 76.73%), 0.81% (78.03% versus 77.22%), and 0.02% (78.03% versus 78.01%), respectively, which are notable margins in terms of mAP. Importantly, ORSR* denotes integrating the ensemble learning strategy with our method, which achieves the best 79.01% mAP. All the comparisons demonstrate that our proposed ORSR achieves a beneficial attempt to unify the exploration of latent supervision, rotation-invariant learning, and GFE into a complete detection framework.

Moreover, we compared our method with many state-of-the-art MS-based OOD approaches. As shown in Table II, all the methods achieve great improvements while using MS training and testing strategy. Specifically, OWSR [31] improves the feature pyramid network (FPN) by exploring segmentation representation which is similar to our proposed ORSR. However, due to the neglect of rotation-invariant learning, OWSR achieves 76.36% mAP which is 4.12% lower than our ORSR. When we adopt an ensemble learning strategy to enhance

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TABLE I

QUANTITATIVE RESULTS OF DIFFERENT SINGLE-SCALE OOD MODELS ON THE DOTA DATASET. RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
FR-O [47]	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
ICN [48]	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CADNet [37]	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
RoI Transformer [28]	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
DRN [49]	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CenterMap [50]	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
SCRDet [51]	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [52]	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
R^3Det [16]	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S^2A-Net [18]	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
OPLD [45]	89.86	82.24	53.02	67.49	74.43	76.96	86.46	90.77	83.28	85.63	58.45	64.84	73.23	67.18	58.75	74.18
AOPG [53]	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77	75.39
DODet [54]	89.61	83.10	51.43	72.02	79.16	81.99	87.71	90.89	86.53	84.56	62.21	65.38	71.98	70.79	61.93	75.89
ReDet [29]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
IBBR [55]	89.55	83.66	54.06	73.93	78.93	83.08	88.29	90.89	86.60	84.80	62.03	65.55	74.16	70.09	58.16	76.25
Oriented R-CNN [41]	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
ORM [56]	89.32	83.59	55.35	74.93	76.15	83.85	87.51	90.69	86.22	85.77	63.83	65.40	77.19	71.23	59.99	76.73
OAN [57]	89.70	84.03	54.61	73.46	79.30	83.27	88.12	90.90	85.22	84.68	62.06	66.87	75.28	70.71	62.75	76.73
FDOL [15]	88.82	84.15	56.25	75.04	74.89	83.88	88.14	90.91	87.97	85.42	59.44	66.36	77.05	74.11	65.89	77.22
SFFD [19]	87.91	83.19	53.87	78.37	73.90	85.30	88.58	90.81	87.67	87.07	66.03	62.53	77.89	76.46	70.60	78.01
ORSR (Ours)	89.53	85.04	57.20	75.05	77.59	83.90	87.80	90.78	88.87	86.74	63.65	66.30	78.05	70.81	69.07	**78.03**
ORSR* (Ours)	88.29	84.38	59.22	80.29	76.74	85.64	88.52	90.89	88.88	86.68	68.62	65.36	78.29	76.84	66.54	**79.01**

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT MS OOD MODELS ON THE DOTA DATASET. RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
FR-O [47]	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
DRN [49]	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
FAOD [52]	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
Gliding Vertex [17]	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
CenterMap [50]	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
OWSR [31]	90.41	85.21	55.00	78.27	76.19	72.19	82.14	90.70	87.22	86.87	66.62	68.43	75.43	72.70	57.99	76.36
OPLD [45]	89.37	85.82	54.10	79.58	75.00	75.13	86.92	90.88	86.42	86.62	62.46	68.41	73.98	68.11	63.69	76.43
S^2A-Net [18]	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
ReDet [29]	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
GWD [58]	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92	80.23
SFFD [19]	89.12	82.50	61.99	81.19	77.85	86.20	88.20	90.87	88.02	84.38	68.42	68.02	84.44	81.00	72.51	80.31
DEA-Net [59]	89.92	83.84	59.65	79.88	80.11	87.96	88.17	90.31	88.93	88.46	68.93	65.94	78.04	79.69	75.78	80.37
DODet [54]	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47	80.62
AOPG [53]	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10	80.66
ORSR (Ours)	88.92	82.19	62.40	80.83	79.22	86.18	88.34	90.07	88.28	86.87	69.43	65.88	83.41	82.22	73.00	**80.48**
ORSR* (Ours)	89.77	84.22	61.49	82.51	78.76	87.11	89.45	91.86	88.67	86.77	72.68	67.58	79.43	78.74	72.68	**80.78**

our method, ORSR* achieves the best 80.78% mAP, which is 0.12% higher than the best competitor AOPG [53].

We also illustrate the visualization comparisons of baseline and our proposed ORSR in Fig. 3. It can be clearly seen that the left visualization of the compared pairs, baseline approach, missing-detect many small oriented instances such as small vehicles. Meanwhile, the baseline method suffers from distinguishing similar instances such as tennis courts and

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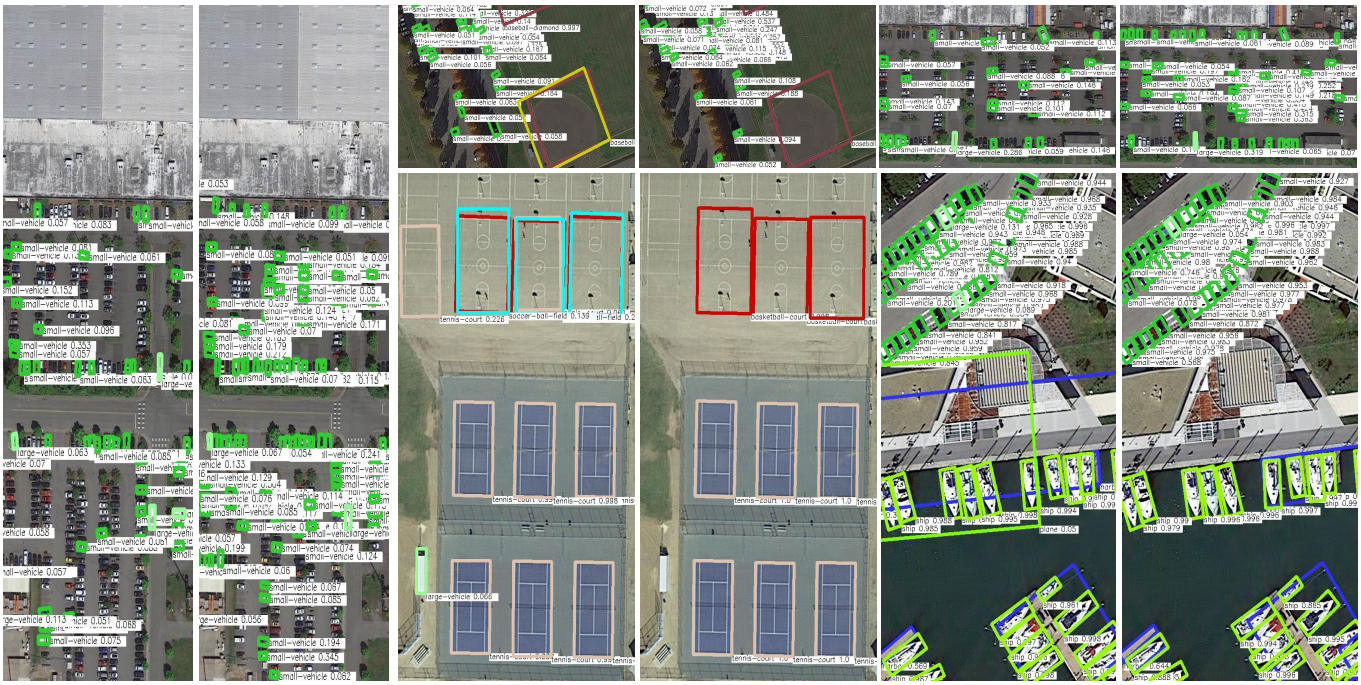


Fig. 3. Visualization of different methods on the DOTA dataset for OOD. The left image of the compared pairs indicates the results of the baseline approach and the right one denotes the results of our proposed ORSR.

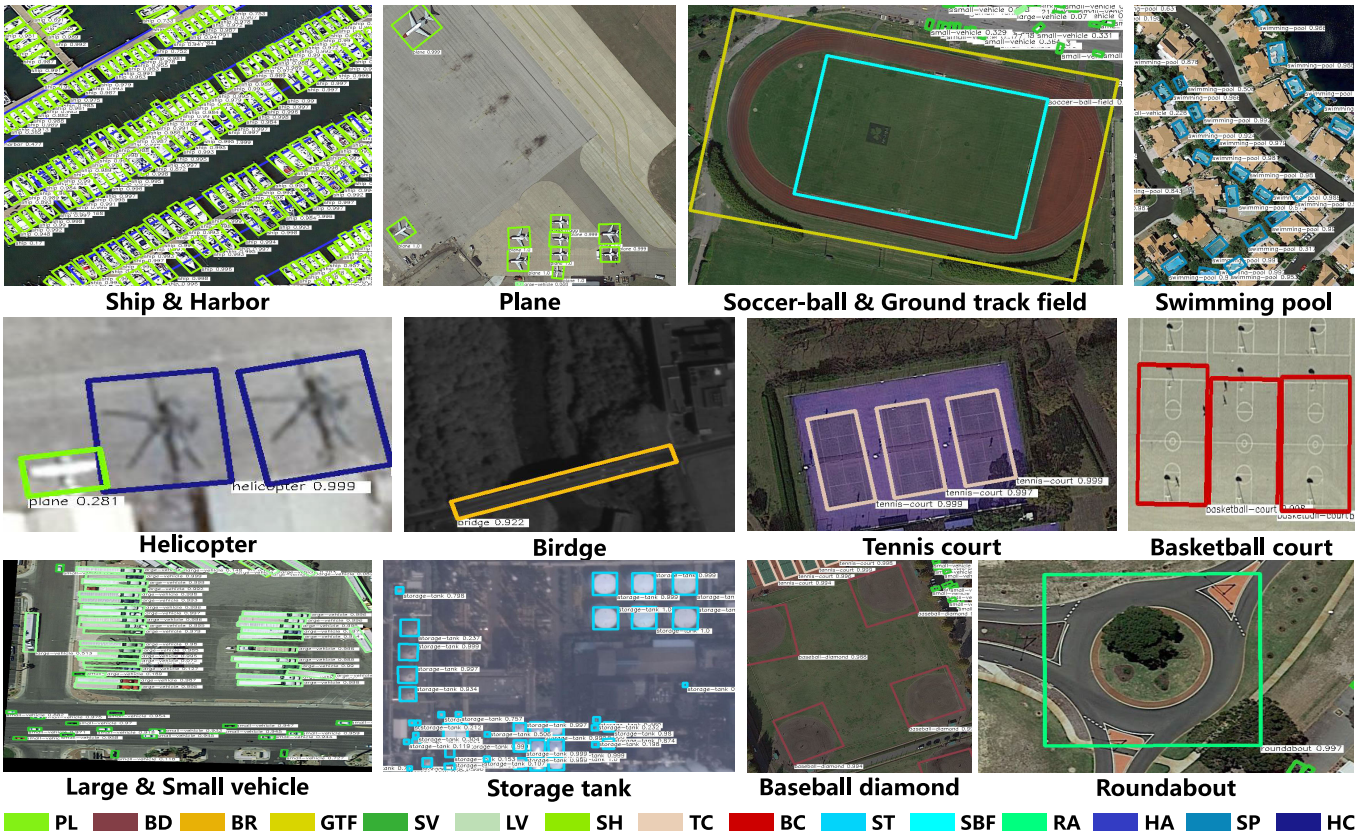


Fig. 4. Visualization results for OOD on the test set of the DOTA dataset.

basketball courts, soccer ball fields, and ground track fields. On the contrary, our proposed ORSR notably detects more objects and successfully eliminates the problem of instance ambiguity. As illustrated in Fig. 4, ORSR accurately classifies different categories and achieves excellent performance in locating small objects, such as vehicles and ships.

2) Comparisons on DIOR-R: Table III lists the results of different OOD methods on the DIOR-R dataset. **

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TABLE III

QUANTITATIVE RESULTS OF DIFFERENT OOD MODELS ON THE DIOR-R DATASET. THE RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED. ALL THE COMPARED METHODS USE 800 × 800 PIXELS OF IMAGES AS INPUT. ** DENOTES THE RESULTS OF ADOPTING THE MS STRATEGY DURING TESTING

Method	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
RTN-O [61]	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
FR-O [47]	62.79	26.80	71.22	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
Gliding Vertex [17]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
OPLD [45]	61.56	38.26	70.29	85.30	42.57	67.33	30.06	77.00	69.23	68.64	83.57	46.69	52.44	81.45	74.66	71.83	80.59	56.54	41.18	63.01	63.11
RoI Transformer [28]	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52	63.87
IBBR [55]	63.22	41.39	71.97	88.55	41.23	72.63	28.82	78.90	69.00	70.07	83.01	47.83	55.54	81.23	72.15	62.66	89.05	58.09	43.38	65.36	64.20
ReDet [29]	77.15	43.68	71.66	87.41	43.70	73.40	34.55	68.55	79.33	69.57	76.23	45.49	59.02	81.24	64.46	69.31	81.42	49.18	48.24	65.20	64.24
AOPG [53]	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41
DODet [54]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10
ORSR (Ours)	63.16	52.24	71.54	89.44	39.47	80.72	34.19	71.13	79.36	70.94	81.68	48.04	57.55	80.76	65.56	61.46	88.85	56.62	46.23	64.56	**65.18**
ORSR** (Ours)	63.37	55.45	72.06	90.11	52.93	81.39	37.30	80.08	89.29	76.26	82.98	54.89	60.78	89.67	70.83	70.54	89.97	62.33	52.04	66.44	**69.94**

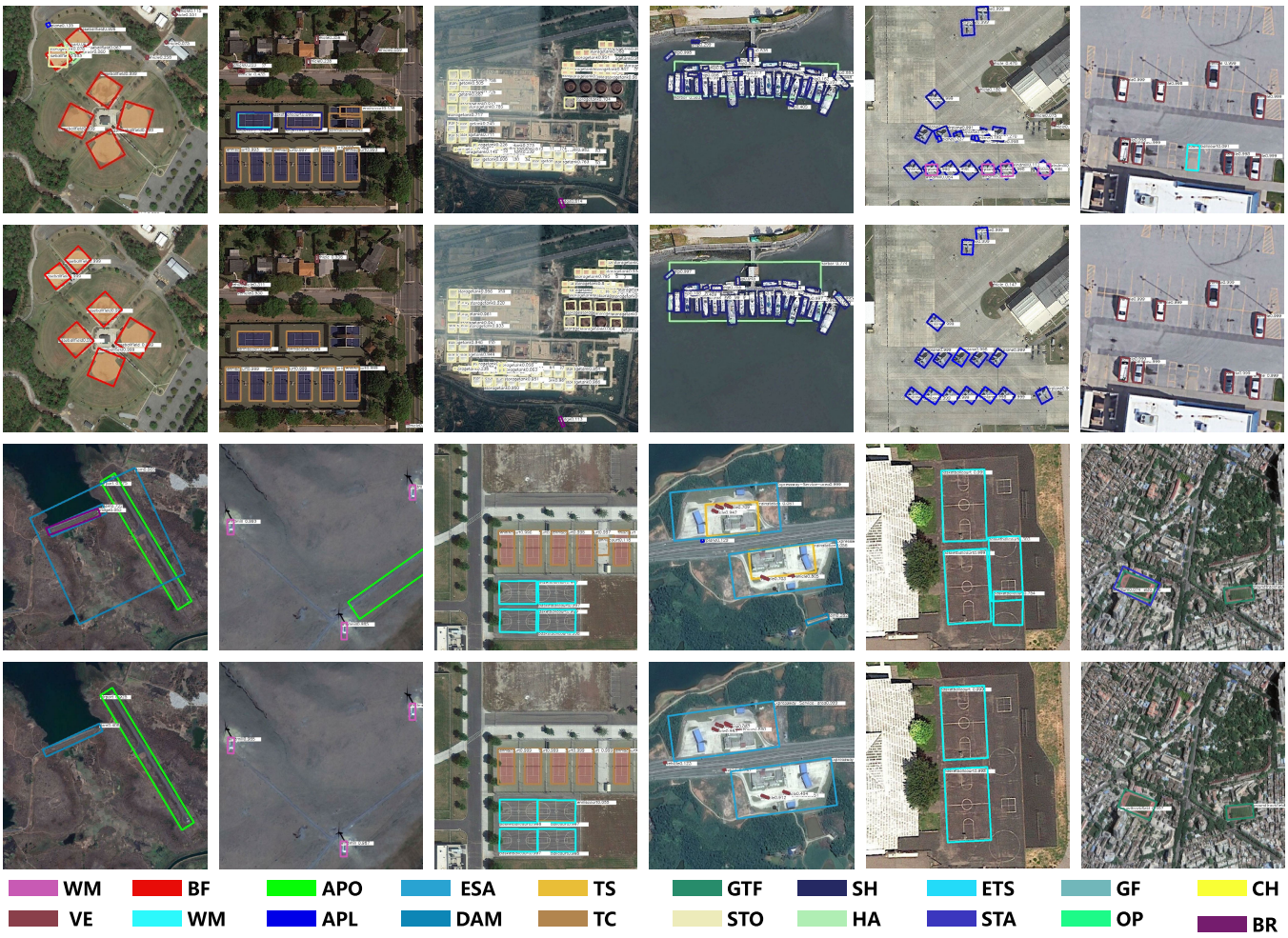


Fig. 5. Visualization comparisons on the DIOR-R dataset for OOD. The first and third rows indicate the detection results of the baseline approaches. The second and fourth rows denote the detection results of our proposed ORSR.

denotes the detection results achieved by adopting the MS strategy during testing. We compare our proposed ORSR with nine recent approaches. Early methods FR-O [47] and RTN-O [61] are all constructed by modifying the corresponding horizontal object detection frameworks to locate different oriented instances, which achieve 57.55% and 59.54% mAP, respectively. Although the aforementioned approaches obtain inferior detection performance, they bring great inspiration for the following works. Gliding Vertex [17] and RoI Transformer [28] are devoted to transferring the horizontal proposals into the oriented ones via predicting the angle of each bounding box. However, due to the limited ability to model rotation-invariant features, the original CNN finds it difficult to obtain the reliably predicted angles of the oriented objects. Considering that the generation of anchor boxes requires a lot of prior knowledge, which is difficult to obtain.

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TABLE IV
RESULTS OF DIFFERENT STATE-OF-THE-ART METHODS IN THE TEST SET OF THE HRSC2016 DATASET

Methods	Backbone	mAP (07)	mAP (12)
RC2 [64]	–	75.70	–
RRPN [43]	R101	79.08	85.64
RoI Transformer [28]	R101	86.20	–
Gliding Vertex [17]	R101	88.20	–
PIoU [65]	DLA-34	89.20	–
DRN [49]	H-34	–	92.70
R^3Det [16]	R101	89.26	96.01
DAL [66]	R101	89.77	–
S^2A-Net [18]	R101	90.17	95.01
AOPG [53]	R101	90.34	96.22
Oriented R-CNN [41]	R50	90.40	96.50
ReDet [29]	ReR50	90.46	97.63
Oriented R-CNN [41]	R101	90.50	97.60
IBBR [55]	R101	90.52	96.64
SFFD [19]	ReR50	90.52	97.68
ORSR (Ours)	ReR50	**90.58**	**97.73**

AOPG [53] proposes an anchor-free detection network and achieves 64.41% mAP, which is 0.20% higher than IBBR [55]. The best competitor DODet [54] focuses on alleviating the misalignment between the classification and localization tasks, which achieves 65.10% mAP. Importantly, the proposed ORSR attains the best 65.18% mAP, which achieves 0.08% and 0.77% gains compared to AOPG and DODet, respectively. We suppose these attributes to the fact that our ORSR emphasizes the representations of image features via the constructed pseudo segmentation branch and also models the rotation-invariant features to perceive different oriented instances in RSIs. Importantly, the size of input images is set to 800 × 800 for all previous methods. Here, we also report the detection performance of our ORSR after adopting the MS testing strategy. The constructed ORSR** achieves 69.94% mAP, which attains significant detection gains compared to all the aforementioned approaches. Fig. 5 illustrates the visualization comparisons of different methods on the DIOR-R dataset. The first and third rows denote the detection results of the baseline method, and the second and fourth rows indicate the detection visualizations of our ORSR. It can be clearly seen that our approach yields more accurate bounding boxes, which also addresses the problem of instance ambiguity for OOD in RSIs.

3) Comparisons on HRSC2016: Table IV lists the results of different OOD methods on the HRSC2016 dataset. We compare our proposed ORSR with 15 recent approaches. Under VOC 2007 and 2012 criteria [13], AOPG [53] follows an anchor-free detection pipeline and achieves 90.34% and 96.22% mAP, respectively. After using a stronger feature extractor, the Oriented R-CNN with ResNet101 backbone network achieves 0.12% and 0.14% mAP gains compared to Oriented R-CNN with ResNet50 backbone, in terms of VOC 2007 and 2012 criteria. The best competitor SFFD [19] attains 90.52% and 97.68% mAP, which is 0.02% and 0.08% higher than Oriented R-CNN, respectively. Importantly, our ORSR achieves the best 90.58% and 97.73% mAP in VOC 2007 and 2012 criteria, which is 0.06% and 0.05% higher than SFFD. The aforementioned performance demonstrates that exploring object-wise rotation invariant semantic representation poses great potential for OOD in RSIs. Fig. 6 illustrates many detection results on HRSC2016 dataset. We can clearly see that the baseline method misses some ships that are extremely long and narrow or close to the image edge. Importantly, our proposed ORSR achieves more accurate OBBs. All the aforementioned experimental results demonstrate the effectiveness of our proposed ORSR.

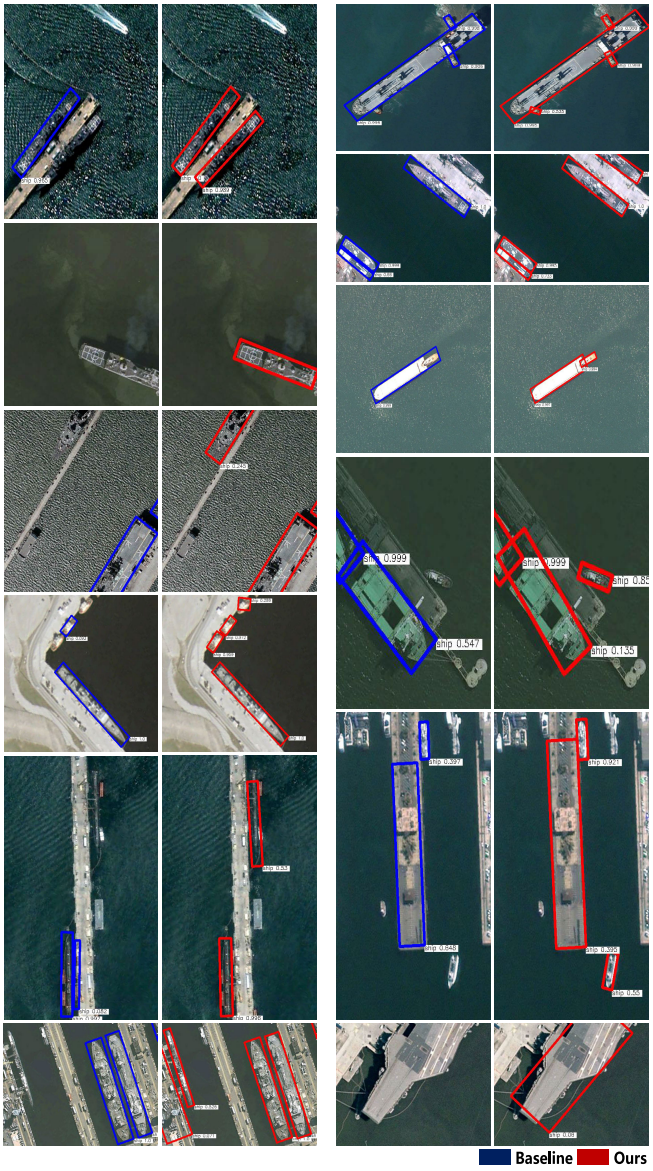


Fig. 6. Visualization of different methods on the HRSC2016 dataset for OOD. The left image of the compared pairs indicates the results of the baseline approach and the right one denotes the results of our proposed ORSR.

D. Ablation Studies

We carry out ablation studies on the DOTA dataset to evaluate: 1) the effect of our proposed segmentation branch; 2) the

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TABLE V
QUANTITATIVE RESULTS MAP OF BASELINE METHODS AND SEGMENTATION SUPERVISION ENHANCED BASELINE METHOD ON THE DOTA DATASET

Baseline	Segmentation Branch	Rotation	$\mathcal{L}_{seg}$	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
OPLD [45] + R50				88.37	83.29	51.92	65.05	70.53	79.71	78.26	91.78	85.81	85.26	60.81	65.83	70.23	67.77	65.10	73.98
	✓		✓	89.86	82.24	53.02	66.49	75.43	76.96	86.46	90.77	83.28	85.63	58.45	64.84	73.23	67.18	58.75	74.17
	✓	✓	✓	89.33	76.79	52.36	66.99	76.65	83.09	89.24	90.84	83.27	85.28	57.04	59.41	74.44	67.52	67.57	**74.65**
OPLD [45] + R101				89.86	82.24	53.02	67.49	74.43	76.96	86.46	90.77	83.28	85.63	58.45	64.84	73.23	67.18	58.75	74.18
	✓		✓	89.38	82.75	54.69	70.54	74.96	75.25	86.16	89.34	81.62	84.82	56.12	63.91	73.80	67.53	62.38	74.22
	✓	✓	✓	90.20	84.73	54.15	70.53	77.80	75.32	85.63	90.85	83.00	87.82	58.27	64.41	73.67	67.43	62.24	**75.07**
ReDet [29] + ReR50				88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
	✓		✓	88.92	83.11	59.05	77.55	76.07	84.09	88.05	90.82	87.65	84.97	60.40	63.41	77.13	70.56	57.37	76.61
	✓	✓	✓	88.69	84.44	55.05	76.64	77.40	83.76	88.03	92.87	89.22	86.98	60.84	68.92	77.06	71.11	59.29	**77.35**

effect of our proposed rotation-invariant learning branch; 3) the effect of our proposed GFE branch; and 4) the computation efficiency of different detection algorithms.

1) Effect of Segmentation Branch: Here, we separate our segmentation branch from ORSR and denote baseline with extra segmentation supervision as "method + segmentation branch + $\mathcal{L}_{\text{seg}}$." Moreover, after introducing rotation augmentation, the baseline with rotated segmentation supervision is named as "method + segmentation branch + rotation + $\mathcal{L}_{\text{seg}}$." As listed in Table V, the original OPLD [45] with ResNet50 feature extractor achieves 73.98% mAP. After introducing extra segmentation supervision (without rotation), our segmentation branch improves the performance of the base OPLD from 73.98% to 74.17% mAP, which achieves 0.19% gains. Importantly, the best performance of 74.65% is attained by further integrating rotation augmentation with segmentation branch-enhanced OPLD. When we use a stronger backbone (ResNet101), the above-mentioned three contrasted methods achieve superior detection performance. ReDet incorporates group convolution [29] into the object detector, which enables to model of rotation-invariant features to a certain extent. Thus, the original ReDet is superior to OPLD and achieves 76.25% mAP. Surprisingly, our proposed extra segmentation supervision still boosts ReDet by 0.36% and 1.10% gains. The aforementioned results demonstrate the considerable potential of exploring latent segmentation supervisions for OOD.

2) Effect of Rotation-Invariant Learning: The premise of rotation-invariant learning is to extract features with their rotated variants. Thus, baseline methods compared in this ablation study are segmentation supervision and rotation augmentation enhanced OPLD + R50, OPLD + R101, and ReDet + ReR50, which achieves 74.65%, 75.07%, and 77.35% mAP, respectively. Here, we denote the aforementioned approaches as "method + Segmentation + Rotation + $\mathcal{L}_{\text{seg}}$." Considering that segmentation predictions remain the same affine transformation before and after rotation, the constructed similarity loss $\mathcal{L}_{\text{sim}}$ aims to encourage the network to model the rotation-invariant features. As listed in Table VI, the $\mathcal{L}_{\text{sim}}$ constrained OPLD achieves 0.18% and 0.13% gains in terms of different backbone networks, which demonstrates that our proposed similarity loss function can assist the detection framework to model the rotation-invariant features. Similarly, experimental comparisons on the ReDet algorithm also prove our analysis. The best performance is still achieved by integrating $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{sim}}$ into the original ReDet, which achieves 77.63% mAP.

TABLE VI
QUANTITATIVE RESULTS MAP OF BASELINE METHODS AND ROTATION-INVARIANT LEARNING ENHANCED BASELINE ON DOTA

Baseline	Segmentation + Rotation + $\mathcal{L}_{seg}$	Rotation-Invariant Learning ($\mathcal{L}_{sim}$)	mAP
OPLD [45] + R50	✓		74.65
	✓	✓	**74.83**
OPLD [45] + R101	✓		75.07
	✓	✓	**75.20**
ReDet [29] + ReR50	✓		77.35
	✓	✓	**77.63**

3) Effect of GFE: Our proposed GFE branch is based on the segmentation module. Therefore, we construct two variants of GFE as "GFE + $\mathcal{L}_{\text{seg}}$" and "GFE + $\mathcal{L}_{\text{con}}$." The latter one denotes our feature enhancement branch, which is supervised by the constructed consistency loss function. Moreover, compared to the common spatial attention mechanism, which tunes the corresponding parameters by target task loss function (e.g., classification and regression loss for object detection), our proposed GFE adopts both consistency loss (our constructed) and classification and regression losses to optimize the corresponding detection algorithm. To verify the effectiveness of our method, we further compared the proposed GFE with two popular spatial attention modules: the convolutional block attention module (CBAM) [67] and self-attention (SE) [68]. As listed in Table VII, the CBAM enhanced OPLD with R50 and R101 achieves 0.34% and 0.39% mAP gains, respectively. However, the CBAM uses a 3×3 convolution kernel to produce attention masks, which inevitably overlooks some small instances in deep-level feature maps. On the contrary, SE discriminates the interaction relationships pixel by pixel, which is more suitable for the requirement of small object detection. Thus, the SE-enhanced OPLD with R50 and R101 achieves 74.59% and 74.95% mAP, which is 0.27% and 0.38% higher than CBAM, respectively. However, due to a lack of rotation-invariant learning, SE suffers from perceiving oriented instances in RSIs. The best performance is still achieved by the proposed "GFE + $\mathcal{L}_{\text{con}}$." Our GFE attains 1.04%, 1.20%,

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TABLE VII
QUANTITATIVE RESULTS MAP OF THE BASELINE METHOD AND THE GFE BRANCH ENHANCED BASELINE METHOD ON THE DOTA DATASET

Baseline	Backbone	Enhancer	mAP
OPLD [45]	R50	—	74.32
		CBAM [67]	74.32
		SE [68]	74.59
		GFE + $\mathcal{L}_{seg}$ (Ours)	74.87
		GFE + $\mathcal{L}_{con}$ (Ours)	**75.02**
OPLD [45]	R101	—	74.17
		CBAM [67]	74.57
		SE [68]	74.95
		GFE + $\mathcal{L}_{seg}$ (Ours)	75.16
		GFE + $\mathcal{L}_{con}$ (Ours)	**75.38**
ReDet [29]	ReR50	—	76.25
		CBAM [67]	76.74
		SE [68]	76.90
		GFE + $\mathcal{L}_{seg}$ (Ours)	77.57
		GFE + $\mathcal{L}_{con}$ (Ours)	**78.03**

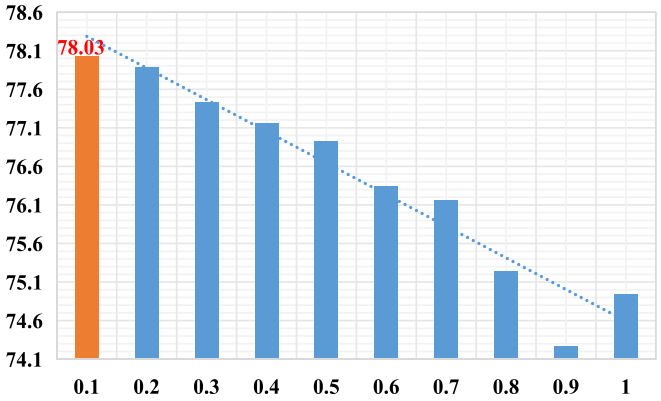


Fig. 7. Influence of λ_3 in our complete loss function defined in (14).

and 1.78% mAP gains compared to the base OPLD with R50, OPLD with R101, and ReDet with ReR50, respectively. All the comparisons demonstrate that our proposed ORSR achieves a beneficial attempt to unify the exploration of latent supervision, rotation-invariant learning, and GFE into a complete detection framework.

From the perspective of multitask learning, different tasks converge at different epochs and the complete model is dominated by the task with a large weight factor. Our proposed ORSR aims to introduce a semantic segmentation branch to assist the object detection task (not dominated). As illustrated in Fig. 7, we range λ_3 from 0.1 to 1.0 for evaluating the influence of λ_3 in our proposed consistency loss function $\mathcal{L}_{\text{con}}$. We can clearly see that the best performance of 78.03% is achieved by setting λ_3 to 0.1. As weight factor λ_3 increases, our ORSR is dominated by the constructed rotation-invariant segmentation branch, while ignoring the most important OOD task. The aforementioned results demonstrate that compared to the classification and regression tasks for OOD, our constructed pseudo-segmentation task is more easier. Thus, the weight factor λ_3 in (14) should be lower than λ_1 and λ_2.

TABLE VIII
COMPUTATION EFFICIENCY INCLUDING MAP, FPS, AND PARAMETERS OF DIFFERENT OOD METHODS ON THE DOTA DATASET

Methods	Backbone	mAP	FPS	Parameter (M)
OPLD [45]	R50	73.98	5.7	48.75
S^2A-Net [18]	R50	74.12	16.4	36.47
OPLD [45]	R101	74.18	2.1	60.13
Oriented R-CNN [41]	R50	75.87	15.6	41.37
OAN [57]	R50	76.02	**20.4**	46.22
ReDet [29]	ReR50	76.25	6.1	**31.65**
Oriented R-CNN [41]	RXt50	76.54	13.4	40.84
OAN [57]	RXt50	76.73	17.5	45.69
FDOL [15]	ReR50	77.22	4.2	32.16
SFFD [19]	ReR50	78.01	4.9	67.95
ORSR	ReR50	**78.03**	5.5	31.68

4) Computation Efficiency of Different Methods: Table VIII lists the computation efficiency of different detection methods. Surprisingly, Oriented R-CNN [41] implements lots of excellent OOD methods, and the criterion of parameter are calculated by their official code. OPLD [45] with ResNet101 backbone achieves 74.18% mAP, 60.13 parameters (M), and 2.1 frames/s, which is more complex than OPLD with ResNet50. OAN [57] is a lightweight fully convolutional network, that judges whether each image patch contains an object or not, and greatly reduces the computational overhead of the object detection algorithm. Therefore, OAN not only obtains more precise detection accuracy but also achieves the fastest inference speed. Oriented R-CNN [41] proposes an excellent variant of RPN, which warps the horizontal bounding boxes into the oriented ones in a nearly cost-free manner. ReDet incorporates group convolution [29] into the object detector and greatly reduces the parameter of the original ResNet50 backbone (31.65 M). However, the computing density of group convolution is smaller, which makes it difficult to make full use of the computing units on GPU. FDOL [15] and SFFD [19] introduce the frequency-domain analysis to model orientation information for perceiving oriented objects. However, due to ignoring the co-scheduling of frequency-domain analysis and convolutional operation, the FPS of the aforementioned methods is lower. Compared with the baseline ReDet [29], our ORSR achieves 78.03% mAP, which attains 1.78% gains. Importantly, the constructed segmentation branch only increases the limited parameter (31.68 M and 31.65 M), which also achieves 5.5 frames/s (5.5 versus 6.1 frames/s).

V. CONCLUSION

In this article, we propose an ORSR network for OOD in RSIs. After exploring the segmentation supervisions from the network itself, our ORSR unifies the rotation-invariant learning and guided attention mechanism into an end-to-end detection pipeline. First, the pseudo segmentation annotations are generated by masking the ground truth of rotated

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bounding boxes in images. The produced segmentation labels are used to supervise the feature selection of our proposed segmentation branch. Then, we introduce a top-down feature fusion pipeline to strike the best balance between high-level semantics and detailed spatial (low-level) information. Furthermore, we propose a consistency loss to encourage the segmentation branch to consistently predict the fused features along with its rotated variants. Importantly, the "object" descriptor predicted in the 2-D segmentation masks can act as a spatial attention map to extend the implicit constraint in our proposed semantic segmentation branch. Surprisingly, the separated descriptor can also be supervised by our constructed consistency loss. The proposed ORSR outperforms many state-of-the-art single-scale and MS methods, which demonstrates the effectiveness of exploring the ORSR for OOD in RSIs.

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